Team 7: SensorFlow | سنسورفلو

Catch salt creep before the tap

Use case: salinity alarms at water-reuse and desalination blending points



The problem

Brine carry-over or saline intrusion first appears as a tiny rise in dissolved salt. Today it is often caught at the next scheduled lab test, after the water has already moved on.

Our target

Detect a **0.15 mg/kg** rise on a **500 mg/kg** NaCl background (0.03 %) in a **1 s** window, so the operator pulls a confirmatory grab sample immediately.

Why it matters: SDG 6

6.1 safe drinking water · 6.3 safe water reuse · 6.4 less water dumped because of late detection. Our question: does quantum sensing make that alarm better?

Sources: UN SDG Report 2025; US EPA Online Water Quality Monitoring. The 0.15 mg/kg target is our design choice.

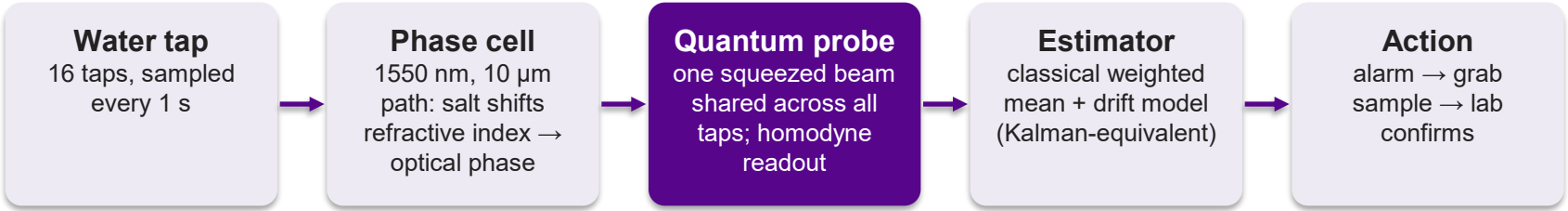


USGS field sampling: real-world monitoring demands

Photos: Duane Lempke / CC0; Barbara Mahler / USGS, public domain

Quantum sensing, classical decisions

End-to-end: water sample → squeezed phase sensing → classical estimator → action



What is quantum

Only the sensing: distributed squeezed light lowers the shot noise on the average of all taps (Zhuang, Zhang & Shapiro 2018; Guo et al. 2020).

Everything after the detector is classical signal processing.

What is not

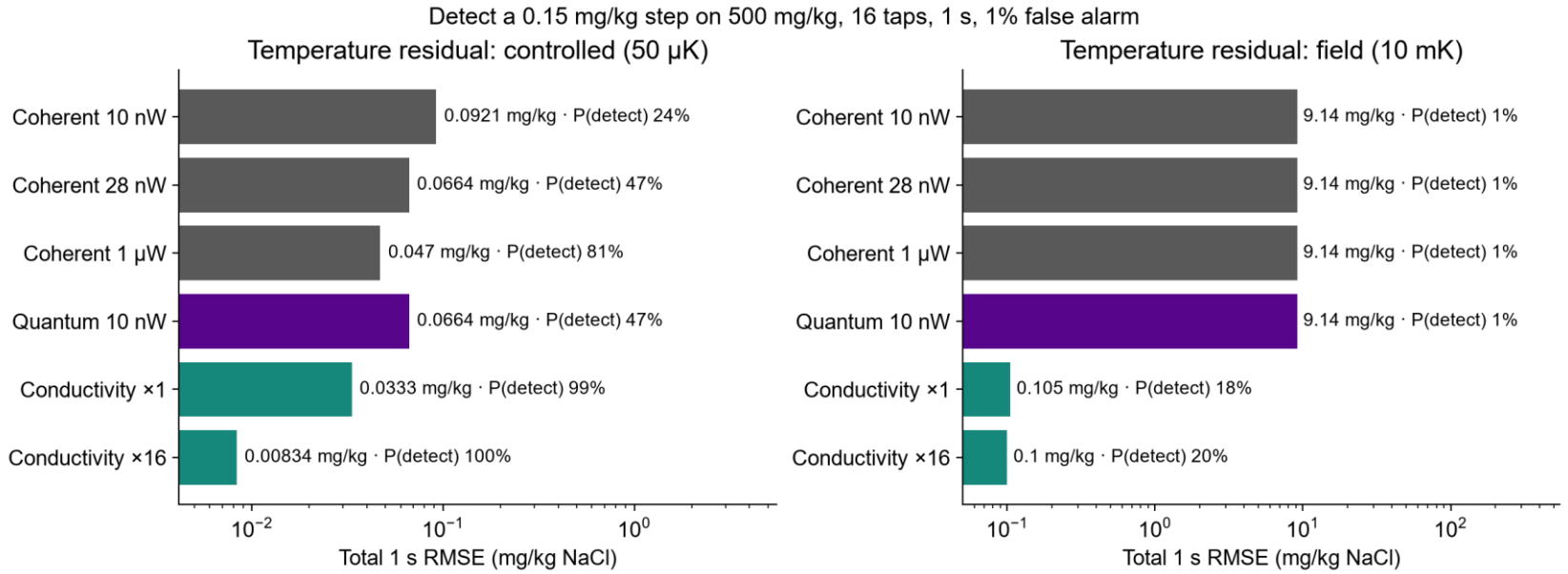
No quantum computing advantage is claimed.

QAOA was tested for choosing tap locations: a classical greedy algorithm is near-optimal and scales to thousands of nodes, so placement moved to backup (B1).

Measured gain is a constant factor, not $\sqrt{N}$. 1.39× lower error for the 16-tap pilot (1.36 dB loss), 1.11× across 64 regional taps. With loss η the ceiling is $1/\sqrt{(1-\eta)}$ = 1.93× (Demkowicz-Dobrzański et al. 2012). See B3.

Is quantum light necessary? Not for salt

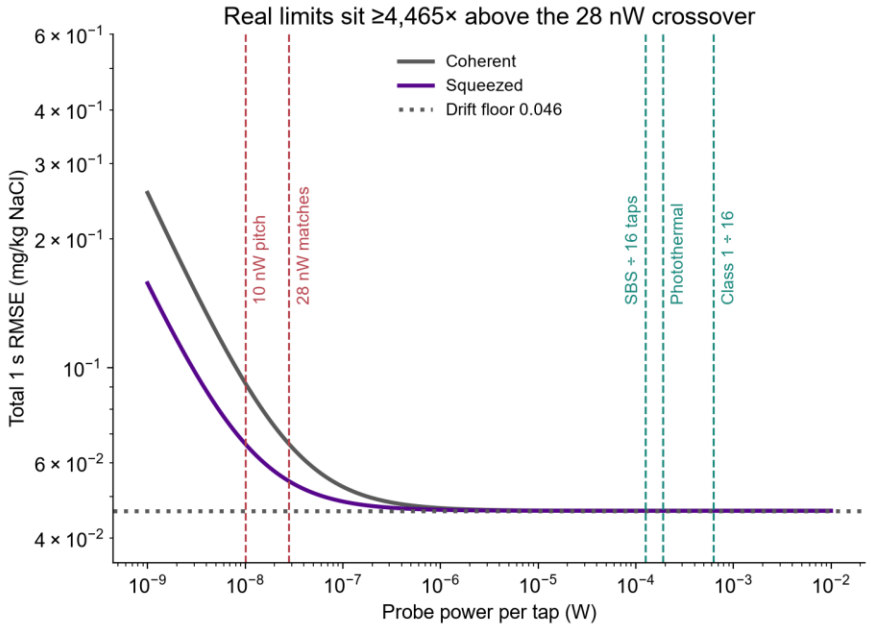
Same 500 mg/kg background, 16 taps, 1 s window, 1 % false alarm per decision



28 nW of ordinary light matches 10 nW of squeezing. A temperature-compensated conductivity probe is 2× better (controlled) and 91× better in field drift: 1 mK looks like 0.92 mg/kg of salt optically, 0.01 mg/kg to conductivity.

What would it take? Power cap and loss budget

Quantum only pays if probe power is genuinely capped — and the squeezing survives the network



No physical cap near 10 nW: fibre SBS ÷ 16 taps allows 125 µW, photothermal noise 189 µW. A real cap needs ~70,000 taps per source.

Element (16-tap pilot)	Loss
Splitter excess (4 stages)	0.20 dB
Fibre (100 m)	0.02 dB
Fibre-to-cell coupling	0.46 dB
Sample cell	0.46 dB
Homodyne detector	0.22 dB
Total, source → detector	**1.36 dB**

Squeezing needed to keep **1.39×**:

9.2 dB entering the network (4.5 dB survives to the detectors)

11.3 dB at the source if 95 % escapes the squeezer; 16.4 dB if 90 %

Gain falls below 1.1× after 5.3 dB of extra loss

Verdict: where quantum light really wins

Squeezing must beat the best classical option at the power the sample allows

	Classical falls short	Power/dose cap binds	Loss ceiling ≥ 1.5×	No time escape	Gain vs best classical
A. Salt pilot (field)	nothing reaches 0.15 mg/kg in field	SBS 125 µW/tap drift/shot noise = 1.28e+04	η = 0.73 → ≤1.93×	slow drift: average longer	0.01×
B. Salt + thermal reference arm	nothing reaches 0.15 mg/kg in field	SBS 125 µW/tap drift/shot noise = 129	η = 0.73 → ≤1.93×	slow drift: average longer	1.00×
C. Plasmonic chip (microcystin)	SPR already 73 ng/L (WHO limit 1 µg/L)	plasmonic heating drift/shot noise = 34.9	η = 0.69 → ≤1.80×	minutes-long binding: average longer	1.00×
D. Live-cell Raman (dose-limited)	fluorescence probes already flag blooms	photodamage (demonstrated) drift/shot noise = 0	η = 0.44 → ≤1.34×	photodamage caps total dose	1.16×

Salt fails. At the real power cap, temperature drift is 129–12,831× the shot noise, so squeezing changes nothing.

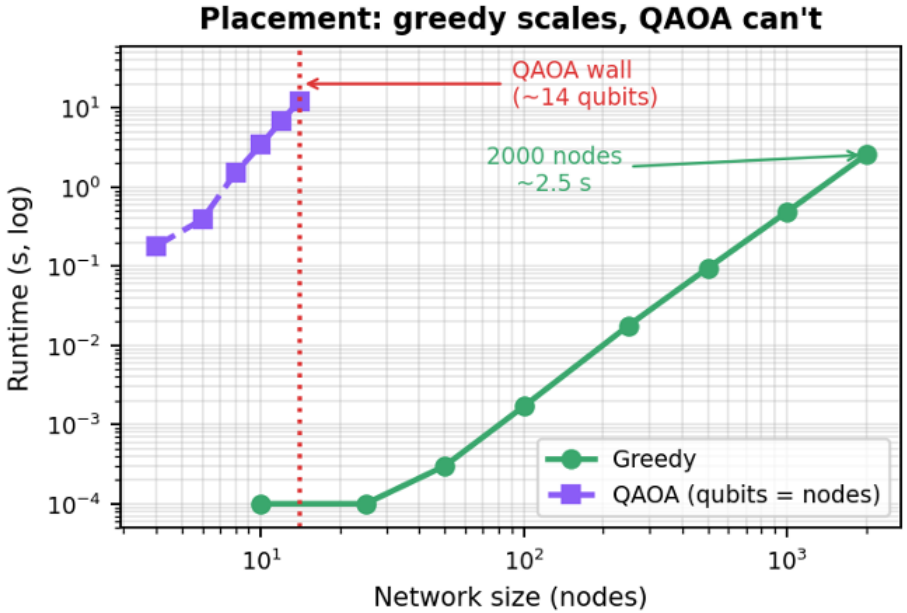
Real advantage needs a dose cap. Live cells limit light by photodamage; squeezing gave 1.16× lower error (Casacio et al. 2021), but 56% optical loss caps it at 1.34×.

Path: 1 find a water decision only a dose-limited measurement can make (e.g. toxin state of live cyanobacteria); 2 tabletop squeezed vs coherent at the photodamage limit; 3 go only if ≥1.2× (today: 1.16×) and classical misses the target.

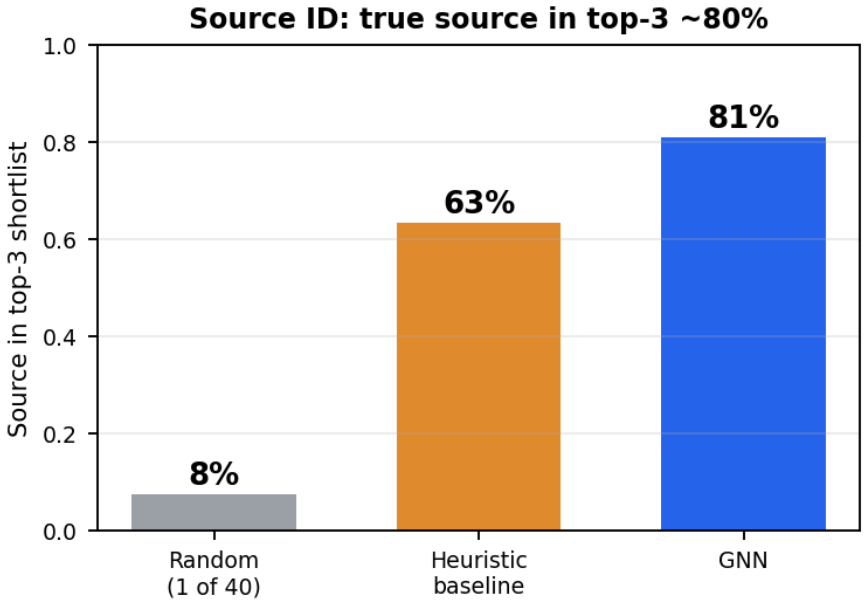
What we deliver: a validated simulator and four-gate test that tells a utility, before it buys a squeezed-light source, whether quantum sensing can beat the sensor it already has.

Backup B1: classical layers — placement and source ID

Decide where to sense (placement) and localize what the regional signal flags (source ID)



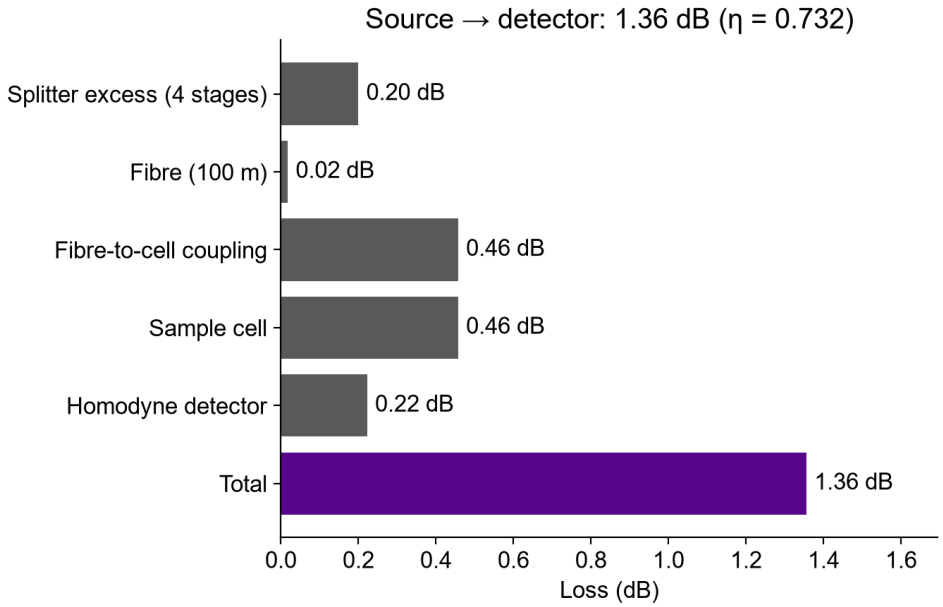
Greedy submodular: provable near-optimal (1−1/e ≈ 63%), scales to thousands of nodes. QAOA (1 qubit/node) matches only on tiny cases, then falls behind and stalls. Placement is a solved classical problem.

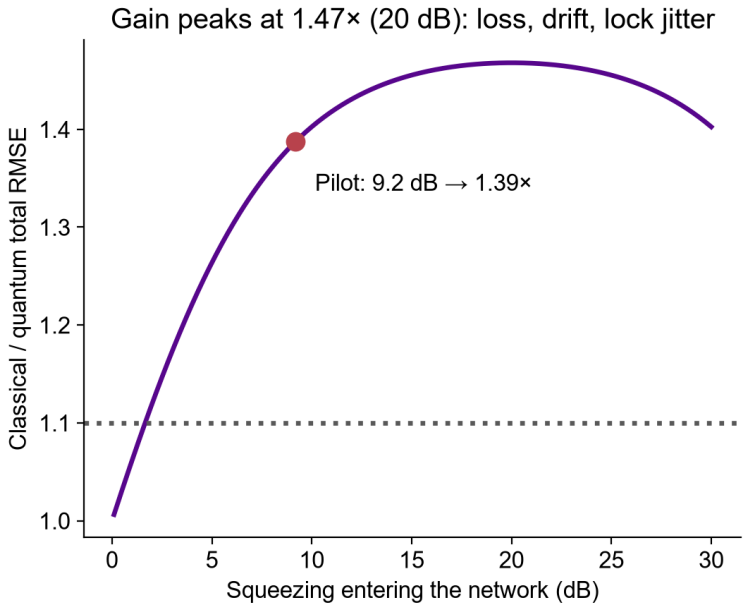


Quantum sensing flags that a region is contaminated (photonic quantum advantage). A GNN recovers likely node source from sensor trip pattern: true source in its top-3 ~80% of the time (vs 7.5% random) on random graphs, degrading gracefully under noise. Further refinement can increase accuracy

Backup B2: loss budget and squeezing sweep

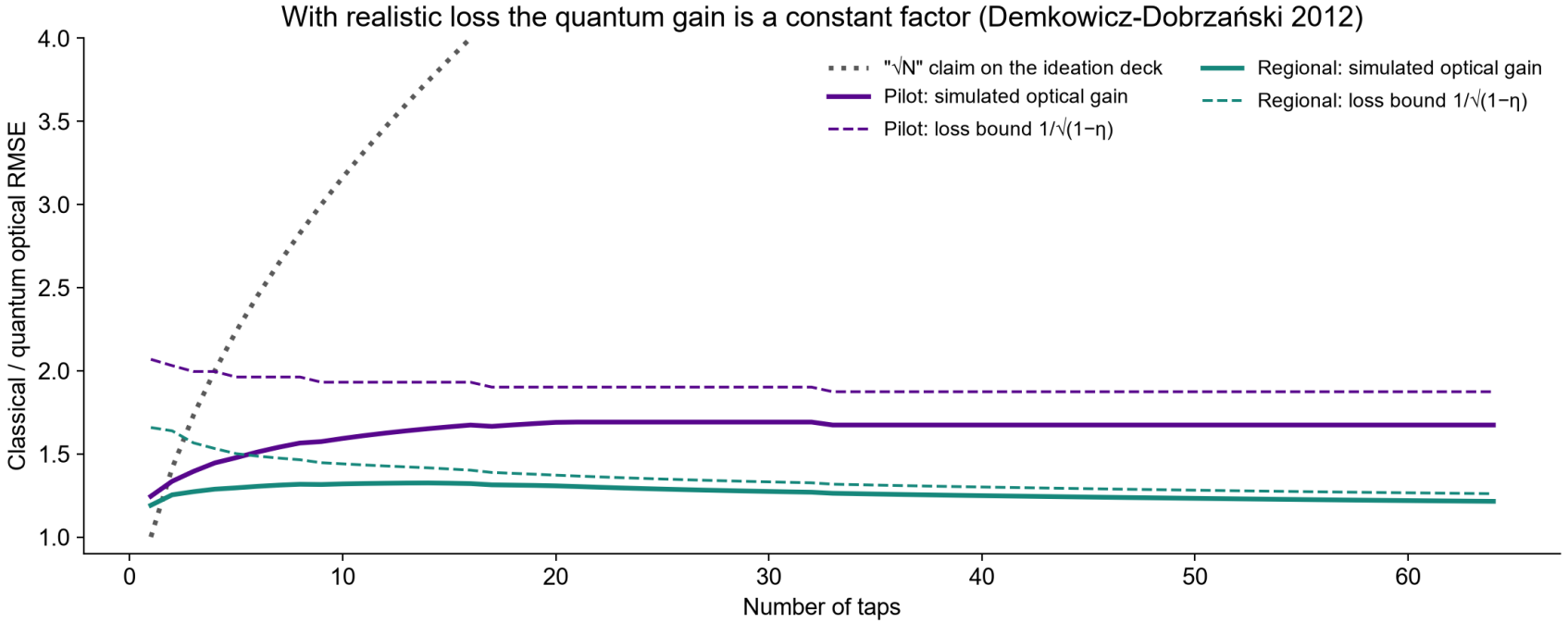
Gain peaks at 1.47× near 20 dB; phase-lock jitter then mixes in anti-squeezing





Backup B3: realised gain vs the √N claim

Loss-limited constant factor 1/√(1−η); Demkowicz-Dobrzański, Kołodyński & Guţă, Nat. Commun. 3, 1063 (2012)

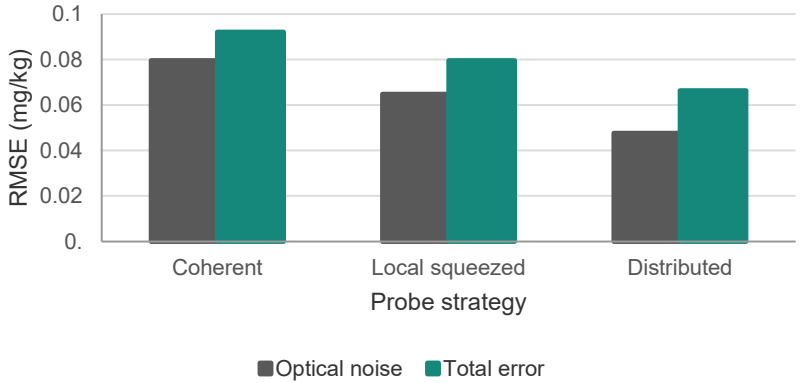


Backup B4: controlled pilot estimation and detection

16 taps • 100 m links • 10 nW/tap • 1 s • Assumed 50 μK common temperature residual

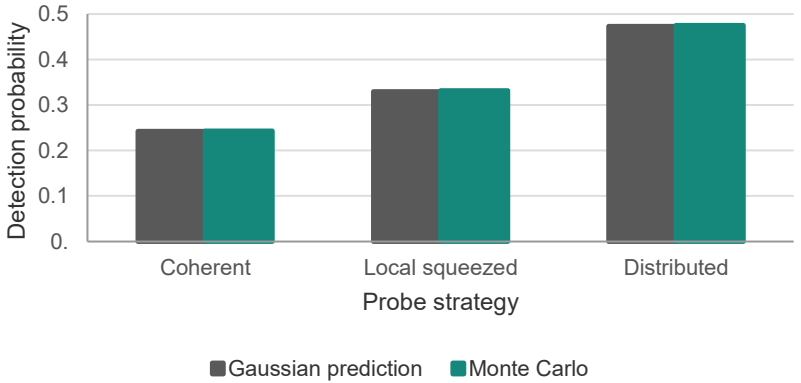
Input: optical budget, 64 locations, salt-change signals; **Compute:** phase sensitivity, optical covariance, mean RMSE, detection probability; **Verification:** 100,000 null/event draws, seeded, 13 model checks

Estimation: 28% lower total RMSE



0.0921 → 0.0664 mg/kg total RMSE. Distributed squeezing also beats independent squeezing by 16.6%.

Detection: 24% → 47% probability



A 0.15 mg/kg mean step, at 1% false alarm per decision. 100,000 null and event trials per strategy.

All results simulated. 1% at 1 Hz means 36 expected flagged windows/hr; need warning system. Analytic error curves, Monte Carlo checks and a synthetic time trace.

Backup B5: sensor and continuous readout

1550 nm light → water-induced phase shift → homodyne → mean estimate

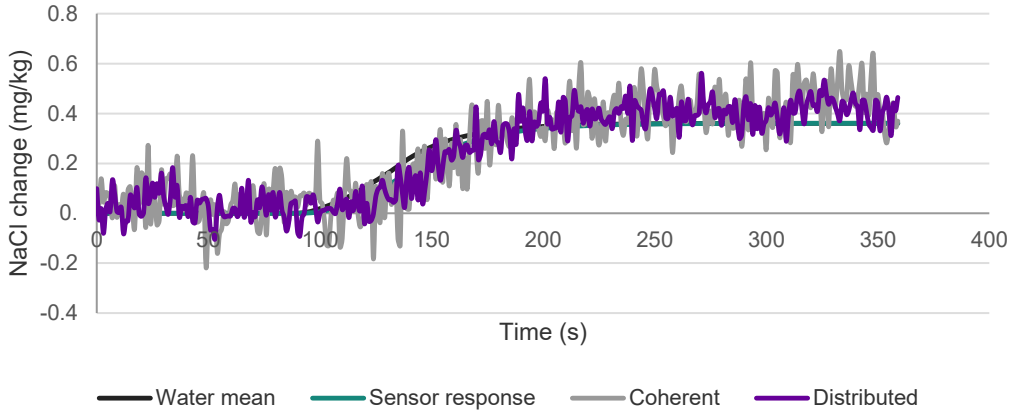
A calibrated optical proxy

NaCl index slope: 1.62×10^{-7} RIU per mg/kg. A proposed 10 µm interaction converts salt changes into phase.

Why shared squeezing?

Distribute one squeezed mode across taps. Negative readout correlations reduce noise in $\hat{q} = \Sigma y_i / M$.

Synthetic event at 16 different taps



Arrivals: 90-135 s; response lag: 10 s. Both strategies share the same drift. Squeezing reduces noise

Classical Python simulator. Proposed quantum sensing hardware. No pollutant-specific identification

Backup B6: assumptions and references

Every number is a model prediction; no hardware was measured

NaCl index slope 1.62×10^{-7} RIU per mg/kg (Saunders et al. 2016); dn/dT -1.49×10^{-4} /K.

10 nW per tap, 10 µm path, 100 m fibre, 16 taps, 1 s window, 50 µK (controlled) or 10 mK (field) common temperature residual.

Water absorption ≈10 cm^{-1} at 1550 nm (Hale & Querry 1973); 5 µm beam radius; 1e-3 laser intensity noise.

Conductivity: 2 µS/cm per mg/kg, 2 %/°C compensation, 0.1 µS/cm resolution, ±0.01 °C sensor (YSI EXO-class specs).

Fibre SBS budget 2 mW per shared trunk; Class 1 eye-safe ~10 mW at 1550 nm.

Zhuang, Zhang & Shapiro, Phys. Rev. A 97, 032329 (2018).

Guo et al., Nat. Phys. 16, 281 (2020).

Demkowicz-Dobrzański, Kołodyński & Guţă, Nat. Commun. 3, 1063 (2012).

Oh et al., Phys. Rev. Research 2, 023030 (2020).

Saunders et al., Appl. Opt. 55, 947 (2016).

Hale & Querry, Appl. Opt. 12, 555 (1973).

YSI EXO sensor specifications; US EPA Online Water Quality Monitoring; UN SDG Report 2025.

Dowran et al., Optica 5, 628 (2018); Casacio et al., Nature 594, 201 (2021).

Code and data: src/judges_response.py, src/advantage_path.py and their results/*.json.

Backup B7: salt verdict, impact and scale

Rigorous negative results are a deliverable: they tell operators what to buy

Verdict

For salinity, a conductivity probe beats our quantum design (2× controlled, 91× in the field).

Squeezing helps only if power per tap is capped below ~30 nW, extra loss stays under ~5 dB and temperature is held to ~0.1 mK.

Impact and scale

At 1 false alarm per week, 16 conductivity probes catch the 0.15 mg/kg step 100% of the time in controlled conditions; the quantum array 0.8%.

Cost, calibration and maintenance favour commercial probes; squeezed sources need lab-grade stabilisation.

Next steps

1 Point the optical sensor at targets conductivity cannot see (non-ionic organics), with a temperature reference arm.

2 Lab pilot: one cell, coherent vs squeezed vs conductivity side by side.

3 Stop rule: if classical meets the operator's alarm target, ship classical.

What we deliver: a validated simulator and feasibility map (power × loss × drift) that tells a utility, before it buys a squeezed-light source, whether quantum sensing will beat the probe it already has.

Backup B8: loss ceiling for any target

Gain over coherent light at the cap with no technical noise; points are the screened candidates

